Computer Science & IT

Discrete Mathematics

Combinatorics

Lecture No. 03

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Recap of Previous Lecture

Topic Method of characteristic roots

Topic Method of undetermined coefficient

Slide

Topics to be Covered

Topic — Generating functions

Topic — Pigeonhole principle

Slide

Generating Function

Topic : Generating function

Generating function is a way of generating an infinite sequence of numbers (a_n) by treating them as the coefficients of a formal power series.

Consider the following sequence,

Seq. : $a_0, a_1, a_2, a_3, a_4, \ldots$

Generating function "Gf(x)" for the given sequence is $= \underbrace{a_0 \cdot x^0 + a_1 \cdot x^1 + a_2 x^2 + a_3 \cdot x^3 + \ldots}$

Solution of this series will be called "generating function in closed form" eg:- Consider the following sequence of real numbers.

$1, 1, 1, 1, \ldots\ldots$

Find generating function for the above sequence in closed form.

Seq = $\underset{a_0}{1}, \underset{a_1}{1}, \underset{a_2}{1}, \underset{a_3}{1}, \ldots\ldots$

Generating function

$$Gf(x) = 1 \cdot x^0 + 1 x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \ldots\ldots$$

$$= \underbrace{1 + x + x^2 + x^3 + \ldots\ldots}$$

it is an infinite GP

if, first term $= a$

& Common ratio $= r$

$\therefore$ Summation $= \dfrac{a}{1-r}$

$$\boxed{Gf(x) = \frac{1}{1-x}}$$

Consider the following sequence of real numbers.

$$a_0,\ a_1,\ a_2,\ a_3, \ldots$$

Generating function

$$G(x) = a_0 x^0 + a_1 \cdot x^1 + a_2 \cdot x^2 + a_3 \cdot x^3 + \ldots$$

$$\boxed{Gf(x) = \sum_{i=0}^{\infty} (a_i) x^i}$$

it is called general term

Binomial Expansion:-

$$(a+b)^n = {}^nC_0 (a)^n (b)^0 + {}^nC_1 (a)^{n-1} (b)^1 + {}^nC_2 (a)^{n-2} (b)^2 + \cdots + {}^nC_r (a)^{n-r} (b)^r + \cdots + {}^nC_n (a)^0 (b)^n$$

$$(b+a)^n = {}^nC_0 \cdot a^0 \cdot b^n + {}^nC_1 \cdot a^1 \cdot (b)^{n-1} + {}^nC_2 \cdot a^2 \cdot b^{n-2} + \cdots + {}^nC_r (a)^r (b)^{n-r} + \cdots + {}^nC_n (a)^n (b)^0$$

$$(1+x)^n = {}^nC_0 (1)^n x^0 + {}^nC_1 (1)^{n-1} (x)^1 + {}^nC_2 (1)^{n-2} (x)^2 + \cdots + {}^nC_r (1)^{n-r} (x)^r + \cdots + {}^nC_n (1)^0 x^n$$

$$\boxed{(1+x)^n = {}^nC_0 \cdot x^0 + {}^nC_1 \cdot x^1 + {}^nC_2 \cdot x^2 + \cdots + {}^nC_r \cdot x^r + \cdots + {}^nC_n \cdot x^n}$$

$$\boxed{(1+x)^n = \sum_{i=0}^{n} \left({}^nC_i\right) x^i}$$

Q: Consider the following seq. of real numbers.

$$ {}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, \ldots, {}^nC_n $$

then what will be the generating function

Soln $Gf(x) = {}^nC_0 \cdot x^0 + {}^nC_1 \cdot x^1 + {}^nC_2 \cdot x^2 + \cdots + {}^nC_n \cdot x^n$

$$ \boxed{Gf(x) = (1+x)^n} $$

eg:- Consider the following sequence of real numbers.

$$\begin{matrix} a_0 & a_1 & a_2 & a_3 & a_n & \\ 1, & 2, & 3, & 4, & 5, & \ldots\ldots \end{matrix}$$

{ it is a sequence of all natural numbers }

Find the generating function for the above sequence

Generating function $Gf(x) = 1\cdot x^0 + 2\cdot x^1 + 3\cdot x^2 + 4\cdot x^3 + \cdots + r\cdot x^{r-1} + (r+1)(x)^r + \cdots$

$r\cdot x^{r-1}$ is actually differentiation of x^r

$$\frac{1}{1-x} = 1\cdot x^0 + 1\cdot x^1 + 1\cdot x^2 + 1\cdot x^3 + \cdots + 1\cdot x^r + \cdots$$

$$\frac{d}{dx}(x^r) = r\cdot x^{r-1}$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^r + x^{r+1} + \cdots$$

$$\frac{d}{dx}(x^{r+1}) = (r+1)\cdot x^r$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\left(1 + x + x^2 + x^3 + \cdots + x^r + x^{r+1} + \cdots\right)$$

$$= 0 + \underbrace{1x^0 + 2\cdot x^1 + 3\cdot x^2 + \cdots + r\cdot x^{r-1} + (r+1)\cdot x^r + \cdots}_{Gf(x)}$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = 0 + Gf(x)$$

$$\boxed{Gf(x) = \frac{d}{dx}\left(\frac{1}{1-x}\right)}$$

$$\boxed{\frac{d}{dx}\left(\frac{P}{Q}\right) = \frac{P'Q - PQ'}{Q^2}}$$

$$Gf(x) = \frac{0\cdot(1-x) - 1\cdot(-1)}{(1-x)^2}$$

$$\boxed{Gf(x) = \frac{1}{(1-x)^2}}$$

eg:- Find the Closed form of generating function.

$Gf(x) = 0 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \dots\dots$

Sequence = 0, 1, 2, 3, - - - - - - { it is a sequence of whole numbers }

(or)

Find the generating function for sequence

0, 1, 2, 3, - - - - - - - { Sequence of all non-negative integers }

eg:- Find the Closed form of generating function.

$Gf(x) = 0 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \cdots$

Sequence = 0, 1, 2, 3, - - - - - $\left\{ \text{it is a sequence of whole number} \right\}$

$Gf(x) = 0 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \cdots + r \cdot (x)^r + \cdots$

We know,

$$\frac{1}{(1-x)^2} = 1 \cdot x^0 + 2 \cdot x^1 + 3 \cdot x^2 + \cdots + r \cdot (x)^{r-1} + \cdots$$

eg:- Find the Closed form of generating function.

$Gf(x) = 0\cdot x^0 + 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \dots$

Sequence $= 0, 1, 2, 3, \dots$ {it is a sequence of whole number}

$$Gf(x) = 0\cdot x^0 + 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \dots + r\cdot(x)^r + \dots$$

We know,

$$\frac{1}{(1-x)^2} = 1\cdot x^0 + 2\cdot x^1 + 3\cdot x^2 + \dots + r\cdot(x)^{r-1} + \dots$$

Multiply LHS & RHS by x

$$\frac{x}{(1-x)^2} = \underbrace{1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \dots + r\cdot x^r + \dots}_{Gf(x)}$$

$$\frac{x}{(1-x)^2} = \underbrace{0\cdot x^0}_{0} + 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \dots + r\cdot x^r + \dots$$

$$\boxed{Gf(x) = \frac{x}{(1-x)^2}}$$

if Seq is $1, 2, 3, 4, 5, - - - - .$ then $Gf(x) = \frac{1}{(1-x)^2}$

Right shift by '1' position

$0, 1, 2, 3, 4 - - - -$ then $Gf(x) = \frac{x}{(1-x)^2}$

Right shift by '2' Position

Right shift by '1' position

$0\ 0\ 1, 2, 3, - - - -$ then $Gf(x) = \frac{x^2}{(1-x)^2} = x \cdot \frac{x}{(1-x)^2}$

* If we want to right shift a sequence by 'K' Position then multiply the sequence by x^K.

And in that case generating function for new sequence will be = x^K * Generating function for original sequence.

Binomial Coefficient

We know, $\displaystyle {}^{n}C_{r} = \frac{n!}{r!\,(n-r)!} = \frac{n\cdot(n-1)\cdot(n-2)\ldots\ldots(n-r+1)\cdot\overbrace{(n-r)(n-r-1)\ldots\ldots3\cdot2\cdot1}^{\cancel{(n-r)!}}}{\left(r\cdot(r-1)(r-2)\ldots\ldots3\cdot2\cdot1\right)\cdot\cancel{(n-r)!}}$

$$\boxed{{}^{n}C_{r} = \frac{(n-0)\cdot(n-1)\cdot(n-2)\ldots\ldots(n-(r-1))}{r!}}$$

from 0 to $(r-1)$ we have

'r' terms $\left\{ \text{we Perform subtraction from 0 to } (r-1) \right\}$

Binomial Coefficient

0 to $r-1$

$(4-1)$

$$^6C_4 = \frac{(6-0)\cdot(6-1)\cdot(6-2)(6-3)}{4!} = \frac{6*5*4*3}{4!}$$

$r = 4$

$(r-1) = 3$

$= 15$ Ans

Extended Binomial Coefficient

Find the value of ${}^{-5}C_3$

$$ {}^{-5}C_3 = \frac{(-5-0)(-5-1)(-5-2)}{3!} $$

$n = -5$

$r = 3$

$\therefore (r-1) = (3-1) = 2$

$$ = \frac{-5 * -6 * -7}{6} = \boxed{-35} $$

Ans

Extended Binomial Coefficient

Find the value of ${}^{-n}C_r$ { where 'n' & 'r' are non-negative integers }

$$ {}^{-n}C_r = \frac{(-n-0)(-n-1)(-n-2)\ldots\ldots(-n-(r-1))}{r!} $$

$$ = \frac{(-n)(-n-1)(-n-2)\ldots\ldots(-n-(r-1))}{r!} $$

$$ = \frac{(-1)^r \cdot \{n \cdot (n+1) \cdot (n+2) \ldots\ldots (n+(r-1))}{r!} $$

$$ = (-1)^r \cdot \frac{\{(n+r-1) \cdot (n+r-2) \ldots\ldots (n+2) \cdot (n+1) \cdot n}{r!} \cdot \frac{(n-1)(n-2)\ldots\ldots 3 \cdot 2 \cdot 1}{(n-1)(n-2)\ldots\ldots 3 \cdot 2 \cdot 1} $$

$$ = (-1)^r \cdot \frac{(n+r-1)!}{r!\,(n-1)!} = \boxed{(-1)^r \cdot {}^{(n+r-1)}C_r} \equiv {}^{-n}C_r $$

Topic : Extended binomial coefficient

$$\boxed{{}^{-n}C_r = (-1)^r \cdot {}^{(n+r-1)}C_r}$$

Where 'n' & 'r' are non-negative integers.

Q:- $\Rightarrow$ Find the $\underbrace{\text{Coefficient of } x^r}_{}$ w.r.t.

↳ {i.e, general term}

following generating functions.

① $Gf(x) = (1+x)^n = {}^nC_0 \cdot x^0 + {}^nC_1 \cdot x^1 + {}^nC_2 \cdot x^2 + \dots + {}^nC_r \cdot x^r + \dots + {}^nC_n x^n$

$$\sum_{i=0}^{n} ({}^nC_i)\, x^i$$

Coefficient of x^r or general term

Coefficient of x^r w.r.t. $(1+x)^n = {}^nC_r$

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② $Gf(x) = (1-x)^{-n}$

$= (1+(-x))^{-n} = {}^{-n}C_0 (-x)^0 + {}^{-n}C_1 (-x)^1 + {}^{-n}C_2 \cdot (-x)^2 + \dots + {}^{-n}C_r \cdot (-x)^r + \dots$

$$= \sum_{r=0}^{\infty} {}^{(n+r-1)}C_r \cdot x^r$$

Coefficient of x^r or general term

$(-1)^r \cdot {}^{(n+r-1)}C_r \cdot (-1)^r \cdot (x)^r$

$\underbrace{(-1)^{2r}}_{1} \cdot {}^{(n+r-1)}C_r \cdot (x)^r$

$|||$

${}^{(n+r-1)}C_r \cdot x^r$

Coefficient of x^r w.r.t. $(1-x)^{-n} = {}^{(n+r-1)}C_r$

③ $Gf(x) = (1-x)^n =$

$$= (1+(-x))^n = {}^nC_0(-x)^0 + {}^nC_1 \cdot (-x)^1 + {}^nC_2 \cdot (-x)^2 + \cdots\cdots + {}^nC_r(-x)^r + \cdots\cdots + {}^nC_n(-x)^n$$

$$\sum_{r=0}^{n} \left\{(-1)^r \cdot {}^nC_r\right\} \cdot x^r$$

$${}^nC_r \cdot (-1)^r \cdot (x)^r$$

$$\left\{(-1)^r \cdot {}^nC_r\right\} \cdot (x)^r$$

Coefficient of x^r w.r.t $(1-x)^n$ is $\left((-1)^r \cdot {}^nC_r\right)$

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④ $Gf(x) = (1+x)^{-n} = {}^{-n}C_0(x)^0 + {}^{-n}C_1 \cdot (x)^1 + {}^{-n}C_2 \cdot (x)^2 + \cdots\cdots + {}^{-n}C_r \cdot (x)^r + \cdots\cdots$

$$(-1)^r \cdot {}^{(n+r-1)}C_r \cdot (x)^r$$

$$\sum_{r=0}^{\infty} \left\{(-1)^r \cdot {}^{(n+r-1)}C_r\right\} \cdot x^r$$

Coefficient of x^r w.r.t. $(1+x)^{-n}$ is $\left\{(-1)^r \cdot {}^{(n+r-1)}C_r\right\}$

⑤ $Gf(x) = (1+ax)^n$

$$= (1+(ax))^n = {}^nC_0 (ax)^0 + {}^nC_1 \cdot (ax)^1 + {}^nC_2 (ax)^2 + \cdots + {}^nC_r \cdot (ax)^r + \cdots + {}^nC_n (ax)^n$$

$${}^nC_r * (a)^r (x)^r$$

$$\{(a)^r \cdot {}^nC_r\} (x)^r$$

$$\sum_{r=0}^{n} \left((a)^r \cdot {}^nC_r\right) (x)^r$$

Coefficient of $(x)^r$ w.r.t. $(1+ax)^n = (a)^r \cdot {}^nC_r$

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⑥ $Gf(x) = (1-ax)^{-n}$

$$= (1+(-ax))^{-n} = {}^{-n}C_0 \cdot (-ax)^0 + {}^{-n}C_1 \cdot (-ax)^1 + {}^{-n}C_2 (-ax)^2 + \cdots + {}^{-n}C_r \cdot (-ax)^r + \cdots$$

$$(-1)^r \cdot {}^{(n+r-1)}C_r * (-1)^r \cdot (a)^r (x)^r$$

$$\{\cancel{(-1)^{2r}} \cdot (a)^r \cdot {}^{(n+r-1)}C_r\} (x)^r$$

$+1$

$$\sum_{r=0}^{\infty} \{a^r \cdot {}^{(n+r-1)}C_r\} \cdot x^r$$

Coefficient of $(x)^r$ w.r.t. $(1-ax)^{-n}$ is $\{(a)^r \cdot {}^{(n+r-1)}C_r\}$

⑦ $Gf(x) = (1-ax)^n = {}^nC_0(-ax)^0 + {}^nC_1(-ax)^1 + {}^nC_2(-ax)^2 + \dots + {}^nC_r(-ax)^r + \dots + {}^nC_n(-ax)^n$

$${}^nC_r \cdot (-1)^r (a)^r (x)^r$$

$$(-1)^r (a)^r \cdot {}^nC_r \cdot x^r$$

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⑧ $Gf(x) = (1+ax)^{-n} = {}^{-n}C_0 \cdot (ax)^0 + {}^{-n}C_1 \cdot (ax)^1 + {}^{-n}C_2 \cdot (ax)^2 + \dots + {}^{-n}C_r \cdot (ax)^r + \dots$

$$(-1)^r \cdot {}^{(n+r-1)}C_r \quad (a)^r \cdot (x)^r$$

$$(-1)^r (a)^r \cdot {}^{(n+r-1)}C_r \cdot x^r$$

⑦ $Gf(x) = (1-ax)^n = {}^nC_0(-ax)^0 + {}^nC_1(-ax)^1 + {}^nC_2(-ax)^2 + \dots + {}^nC_r(-ax)^r + \dots + {}^nC_n(-ax)^n$

Coefficient of $(x)^r$ w.r.t $(1-ax)^n$ is $\{(-1)^r \cdot (a)^r \cdot {}^nC_r\}$

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⑧ $Gf(x) = (1+ax)^{-n} = {}^{-n}C_0 \cdot (ax)^0 + {}^{-n}C_1 \cdot (ax)^1 + {}^{-n}C_2 \cdot (ax)^2 + \dots + {}^{-n}C_r \cdot (ax)^r + \dots$

Coefficient of $(x)^r$ w.r.t $(1+ax)^{-n}$ is $\{(-1)^r \cdot (a)^r \cdot {}^{(n+r-1)}C_r\}$

"Gate-PYQ"

it is generating function of 1, 2, 3, 4, ...

i.e $1\cdot x^0+2\cdot x^1+3\cdot x^2+\cdots+(i+1)\cdot x^i \cdots$

#Q. Let $G(x)=\frac{1}{(1-x)^2}=\sum_{i=0}^{\infty} g(i)\, x^i$, where $|x|<1$. What is $g(i)$?

it is coefficient of $x^i \Rightarrow \therefore (i+1)$

$(1-x)^{-2} \Rightarrow \therefore$ Coefficient of $x^i = {}^{(n+i-1)}C_i = {}^{(2+i-1)}C_i$

$n=2$

$= {}^{(i+1)}C_i = {}^{(i+1)}C_1 = (i+1)$ Ans

${}^nC_r = {}^nC_{n-r}$

"Gate-PYQ"

#Q. Let $G(x) = \frac{1}{(1-x)^2} = \underbrace{\sum_{i=0}^{\infty} g(i)\, x^i}$, where $|x| < 1$. What is $g(i)$?

Coefficient of x^i

we know $\frac{1}{(1-x)^2}$ is the generating function for sequence. $1, 2, 3, 4, \ldots$

i.e. $1.(x)^0 + 2.(x)^1 + 3(x)^1 + \ldots + \underbrace{(i+1)\, x^i}_{} + \ldots$

Coefficient of x^i is $(i+1)$

$\therefore g(i) = (i+1)$

Ans

Coefficient of $(x)^i$ in $\frac{1}{(1-x)^2}$

i.e. Coefficient of $(x)^i$ in $(1-x)^{-2} = {}^{2+i-1}C_i$

$= {}^{(i+1)}C_i = {}^{(i+1)}C_{(\cancel{i}+1-\cancel{i})} = {}^{(i+1)}C_1$

$= (i+1)$

$a_0, a_1, a_2, a_3 \ldots\ldots$

#Q. If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to?

in the Expansion of $\frac{(1+z)}{(1-z)^3}$. a_i is Coefficient of $(z)^i$

Coefficient of $(z)^3$

Coefficient of $(z)^0$

$$\frac{(1+z)}{(1-z)^3} = (1+z)\cdot(1-z)^{-3}$$

Coefficients of Z^i for $i \geq 2$ are all '0'

$$\frac{1+Z}{(1-Z)^3} = (1+Z) \cdot (1-Z)^{-3} = \left(1 \cdot Z^0 + 1 \cdot Z^1\right) \cdot (1-Z)^{-3}$$

Overall co-efficient of Z^0 in the Expansion of $\frac{1+Z}{(1-z)^3}$ = {Coefficient of Z^0 w.r.t $(1+Z)$} * {Coefficient of Z^0 w.r.t. $(1-z)^{-3}$}

Overall co-efficient of Z^3 in the Expansion of $\frac{1+Z}{(1-z)^3}$ = {Coefficient of Z^0 w.r.t. $(1+Z)$ * Coefficient of Z^3 w.r.t. $(1-Z)^{-3}$} + {Coefficient of Z^1 w.r.t. $(1+Z)$ * Coefficient of Z^2 w.r.t. $(1-Z)^{-3}$}

Overall co-efficient of Z^0 in the Expansion of $\dfrac{1+Z}{(1-Z)^3}$ = $\left\{\begin{array}{c}\text{Coefficient of } Z^0 \\ \text{w.r.t } (1+Z)\end{array}\right\} \times \left\{\begin{array}{c}\text{Coefficient of } Z^0 \\ \text{w.r.t } (1-Z)^{-3}\end{array}\right\}$

$$1 \times {}^{3+0-1}C_0$$

$$1 \times {}^{2}C_0 = 1 \times 1 = 1$$

$$\therefore \boxed{a_0 = 1}$$

Overall co-efficient of z^3 in the Expansion of $\frac{1+z}{(1-z)^3}$ = $\left\{\begin{matrix}\text{Coefficient of } z^0 \\ \text{w.r.t.} \\ (1+z)\end{matrix} * \begin{matrix}\text{Coefficient of } z^3 \\ \text{w.r.t.} \\ (1-z)^{-3}\end{matrix}\right\} + \left\{\begin{matrix}\text{Coefficient of } z^1 \\ \text{w.r.t.} \\ (1+z)\end{matrix} * \begin{matrix}\text{Coefficient of } z^2 \\ \text{w.r.t.} \\ (1-z)^{-3}\end{matrix}\right\}$

$$\left\{ '1' * {}^{3+3-1}C_3 \right\} + \left\{ 1 * {}^{3+2-1}C_2 \right\}$$

$$\left(1 * {}^5C_3\right) + \left(1 * {}^4C_2\right)$$

$$10 + 6$$

$$= 16$$

$$\therefore \boxed{a_3 = 16}$$

#Q. If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to?

$$a_3 - a_0 = 16 - 1 = 15$$

Ans

Q:- Find the Coefficien of x^{12} in the expansion of

$$\left(x^3+x^4+x^5+x^6+\cdots\cdots\right)^3$$

$$\left(x^3\left(1+x+x^2+\cdots\cdots\right)\right)^3$$

$$x^9\left(1+x+x^2+\cdots\cdots\right)^3$$

$$x^9\cdot\left(\frac{1}{1-x}\right)^3$$

$$x^9\cdot(1-x)^{-3}$$

Coefficient of x^{12} in $x^9\cdot(1-x)^{-3}$ = Coefficient of x^3 in $(1-x)^{-3}$

$$= {}^{3+3-1}C_3$$

$$= {}^5C_3 = 10$$

Ans

Type: When general term is given, and we need to identify the generating function

#Q. Which one the following is a closed from expression for the generating function of the sequence $\{a_n\}$, where $\underbrace{a_n = 2n + 3}$ for all $n = 0,1,2, \ldots$?

A) $\frac{3}{(1-x)^2}$

B) $\frac{3x}{(1-x)^2}$

C) $\frac{2-x}{(1-x)^2}$

D) $\frac{3-x}{(1-x)^2}$

$$Gf(x) = \sum_{n=0}^{\infty} (a_n) \cdot x^n$$

$$Gf(x) = \sum_{n=0}^{\infty} (2n+3) \cdot x^n$$

$$= \sum_{n=0}^{\infty} (2n) \cdot x^n + \sum_{n=0}^{\infty} (3)(x)^n$$

$$Gf(x) = 2 \cdot \sum_{n=0}^{\infty} (n)(x)^n + 3 \cdot \sum_{n=0}^{\infty} (1) \cdot (x)^n$$

general term $= n$

then $g.f = \dfrac{x}{(1-x)^2}$

general term $= 1$

$\therefore g.f = \dfrac{1}{1-x}$

$$Gf(x) = 2 \cdot \frac{x}{(1-x)^2} + 3 \, \frac{1}{(1-x)}$$

$$= \frac{2x + 3(1-x)}{(1-x)^2} = \boxed{\frac{3-x}{(1-x)^2}}$$

Ans

Type: When general term is given, and we need to identify the generating function

#Q. Which one the following is a closed from expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0,1,2, \ldots$?

A) $\frac{3}{(1-x)^2}$

B) $\frac{3x}{(1-x)^2}$

C) $\frac{2-x}{(1-x)^2}$

D) $\frac{3-x}{(1-x)^2}$

#Q. Find the solution of the recurrence relation using Concept of generating function

$a_k = 3a_{k-1}$, where k= 1, 2, 3, ... and $a_0=2$

$$Gf(x) = \sum_{k=0}^{\infty} (a_k) \cdot x^k \equiv \cancel{a_0 \cdot x^0} + \sum_{k=1}^{\infty} (a_k) \cdot x^k$$

2   1

$$= 2 + \sum_{k=1}^{\infty} (3 \cdot a_{k-1}) \cdot (x)^k$$

$$= 2 + 3 \cdot \sum_{k=1}^{\infty} (a_{k-1}) \cdot (x)^k$$

#Q. Find the solution of the recurrence relation using Concept of generating function

$a_k = 3a_{k-1}$, where k= 1, 2, 3, ... and =2

$$Gf(x) = \sum_{k=0}^{\infty} (a_k) \cdot x^k = 2 + 3 \cdot \sum_{k=1}^{\infty} a_{k-1} \cdot x^k$$

$$= 2 + 3x \sum_{k=1}^{\infty} a_{k-1} \cdot x^{k-1}$$

when 'k' vary from '1' to ∞, then 'k-1' will vary from 0 to ∞

$$Gf(x) = 2 + 3x \underbrace{\sum_{k=0}^{\infty} a_k x^k}_{Gf(x)}$$

$$Gf(x) = 2 + 3x \cdot Gf(x)$$

$$\boxed{Gf(x) = \left(\frac{2}{1-3x}\right)}$$

#Q. Find the solution of the recurrence relation using Concept of generating function

$a_k = 3a_{k-1}$, where k= 1, 2, 3, ... and =2

$$Gf(x) = \sum_{k=0}^{\infty} (a_k) \cdot x^k \equiv \frac{2}{(1-3x)}$$

Coefficient of x^k in the expansion of $\left(\frac{2}{1-3x}\right)$ is our a_k

$$a_k = \text{Coefficient of } x^k \text{ in } 2\cdot(1-3x)^{-1}$$

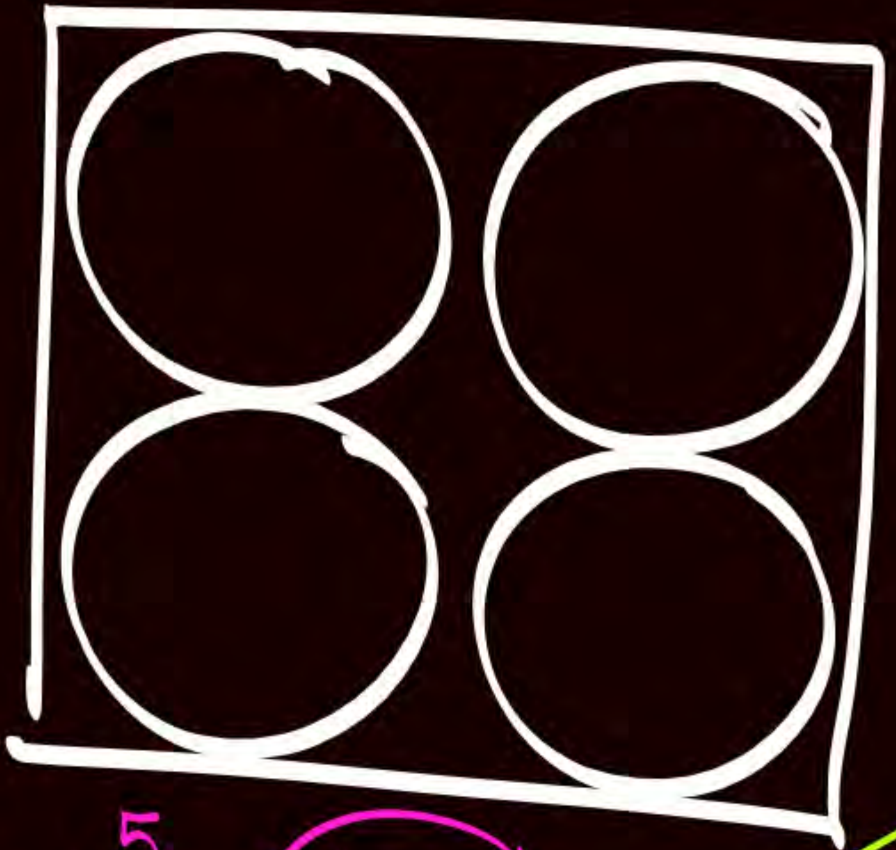
$$= 2 * \left\{\text{Coefficient of } x^k \text{ in } (1-3x)^{-1}\right\}$$

$$= 2 * (3)^k \cdot {}^{1+k-1}C_k = 2\cdot 3^k \cdot \cancel{{}^{k}C_k}^{\,1} = 2\cdot 3^k$$

Pigeonhole Principle

let No. of pigeonholes = 4
&
No. of Pigeons = 5

then Average no. of Pigeons per pigeonhole will be = $\dfrac{\text{No. of Pigeons}}{\text{No. of Pigeonholes}} = \dfrac{5}{4} = 1.25$



Some pigeonhole will Contain at least '2' pigeons

$\lceil 1.25 \rceil = 2$

$\lfloor 1.25 \rfloor = 1$

Some Pigeonhole will Contain at most '1' Pigeons

Topic : Pigeonhole principle

- Let 'A' is the average no. of Pigeons per pigeonhole

$$A = \frac{\text{No. of Pigeons}}{\text{No. of Pigeonholes}}$$

$\therefore$ Some pigeonhole will Contain at least $\lceil A \rceil$ Pigeons

and Some pigeonhole will Contain at most $\lfloor A \rfloor$ Pigeons

→ Let $(n+1)$ Pigeons are distributed in 'n' pigeonholes

$(1 n+1)$

then $A = \frac{n+1}{n} = 1 + \frac{1}{n}$

$\therefore \lceil A \rceil = 2 = (1+1)$

$\lfloor A \rfloor = 1 = (1)$

∴ Some pigeonholes will contain at least '2' Pigeons

& Some Pigeonholes will contain at most '1' Pigeons

→ Let $(2n+1)$ Pigeons are distributed in 'n' pigeonholes

$$A = \frac{2n+1}{n} = 2 + \frac{1}{n}$$

$$\lceil A \rceil = 3 = (2+1)$$

$$\lfloor A \rfloor = 2 = 2$$

∴ Some Pigeonholes will Contain at least '3' Pigeons

& Some Pigeonholes will Contain at most '2' Pigeons

- If $(Kn+1)$ Pigeons are distributed in 'n' pigeonholes, then $A = \dfrac{Kn+1}{n} = K + \dfrac{1}{n} \Rightarrow \lceil A \rceil = K+1$, $\lfloor A \rfloor = K$

- Some Pigeonhole will Contain at least $(K+1)$ Pigeons And Some Pigeonhole will Contain at most $(K+0)$ Pigeons

Q :- Suppose we have 'n' pigeonholes. then what is the minimum number of pigeons required such that

upto $(k-1)n$ pigeons, there may be possibility s.t. every pigeonhole has less than 'k' pigeons

Q. (i) some pigeonhole contains at least 'k' pigeons Ans = $((k-1)n+1)$

Q. (ii) some pigeonhole contains at least $(k+1)$ pigeons Ans = $(k \cdot n+1)$

MSQ

~~#~~Q. Consider a group of 61 students, which of the following is/are true?

Pigeons & '12' Months are '12' Pigeonholes

$$61 = ((5 * 12) + 1)$$

Pigeons = (5 * No. of pigeonholes + 1) → ∴ in some month at least (5+1) students are born & in some month at most '5' students are born

A) At least 6 students are born is same month.

B) At most 6 students are born in same month.

C) At least 5 students are born in same month.

D) At most 5 students are born in same month.

Pigeonholes & Students are Pigeons

#Q. Suppose there are 7 branches in a college and each branch has exactly 50 students, what must be the minimum number of students chosen such that there are at least 10 students from at least one branch?

We need at least '10' students from some branch

$(K+1) \Rightarrow (K=9)$

i.e. we need at least '10' pigeons in at least one Pigeonhole

$\therefore$ Min no of Students selected $= (K \cdot n + 1)$

$= (9 \cdot 7 + 1) = 64$

Upto $9 * 7 = 63$ Students there is a possibility that we have '9' students from each branch

i.e. no '10' students from any branch

but if we select '1' more student {i.e. '64' students} then it is guaranteed that there will be at least '10' students from some branch

#Q. Suppose we have 8 red color balls, 6 blue color balls, 19 green color balls and 10 yellow color balls, then what must be the minimum number of balls chosen such that we have at least 5 balls of same color?

Red	Blue	Green	Yellow	
4 +	4 +	4 +	4	= 16

Upto '16' balls it is Possible that we don't have '5' balls of same Color

So Ans = 16 + 1 = 17

2 mins Summary

Topic — Generating functions

Topic — Pigeonhole principle

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THANK - YOU

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